

UMTS — the mobile part of broadband communications for the next century

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This paper outlines the path from developing broadband ISDN/ATM systems towards the European vision of the next generation mobile system — Universal Mobile Telecommunications System (UMTS).

The capabilities of UMTS in terms of services and features compared to the second generation mobile system GSM, and the network technologies behind N-ISDN and B-ISDN, are discussed with proposals for technical developments to satisfy the UMTS requirements. The UMTS radio interface aspects are described and considerations on the network requirements to support these radio aspects are discussed.

The requirements within the network for control, switching and transport to support the mobility, service and interconnecting network aspects of UMTS are outlined in conjunction with the technical solutions currently under discussion and some latest thoughts on a potential high-level architecture

1. Introduction

This paper discusses the radio and network requirements and developments for the Universal Mobile Telecommunications System (UMTS) and outlines the initial proposals for the network architecture and capabilities of UMTS compared to the global system for mobile communications (GSM) and broadband ISDN (B-ISDN).

Within Europe the mobility aspects of UMTS are building upon GSM, with recent architectural developments within GSM acting as the basis for the mobile evolution path for the future mobile system. The transport requirements must also be addressed and it is here that broadband based upon ATM is the obvious candidate. This will allow the support of the dynamic, variable bit rate and asymmetric communications requirements of UMTS to be provided. The objective is to combine the variable, high bit rate capabilities of ATM with the developing intelligence control of B-ISDN and the high mobility and advanced supplementary service support that GSM offers.

One of the major successes of GSM has been its complete system specification in conjunction with the provision of basic standardised service sets and automated inter-operator roaming. While this has enabled operators to rapidly roll out feature-rich networks, it has left little room for service innovation and differentiation via operator-specific services. The GSM community has addressed this requirement via the development of the GSM CAMEL

(customised applications for mobile network enhanced logic) feature [1]. CAMEL uses evolved IN capabilities, based on the INCS (intelligent network capability set) evolution model, to provide operator-specific services in both the home and roamed to (visited) networks. This is achieved by developing architectural SSP and IN signalling capabilities (GSM CAP protocol [2]) while catering for the inter-operator roaming and mobile communications limitations. The service creation in the CAMEL platform is directly controlled by the home network operator.

Figure 1 illustrates the phase 1 CAMEL architecture and shows the inclusion of the IN capability into the GSM network.

CAMEL provides an evolution step towards the 'virtual home environment' (VHE) feature of UMTS. VHE aims to enable users to access their UMTS services and features in a common configurable manner independent of the environment (e.g. public cellular, business/private, wireless PBX, or home). This capability will ease many of the problems of common service support while roaming between the different public and private networks and accesses (termed access domains or environments) shown in Fig 2. Access to different environments is already being provided within the GSM community by the use of dual/multi-mode GSM, DCS1800 and DECT terminals. UMTS will build upon these facilities to provide the true seamless access and support of multiple environments.

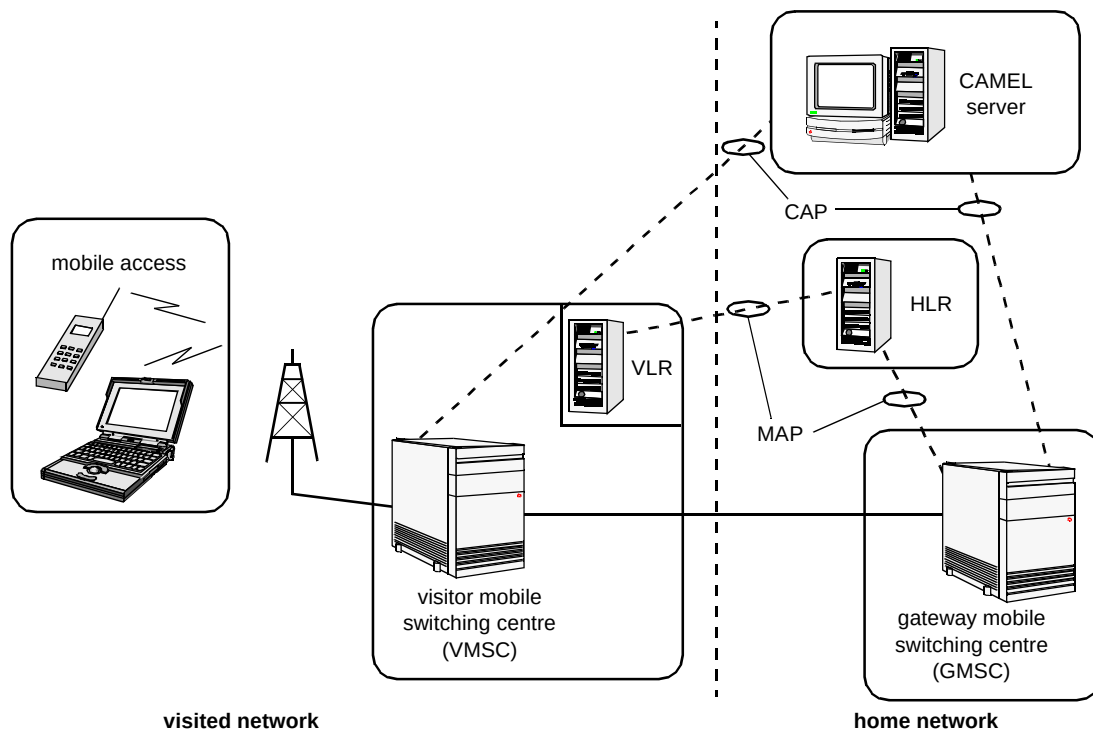


Fig 1 The GSM CAMEL architecture.

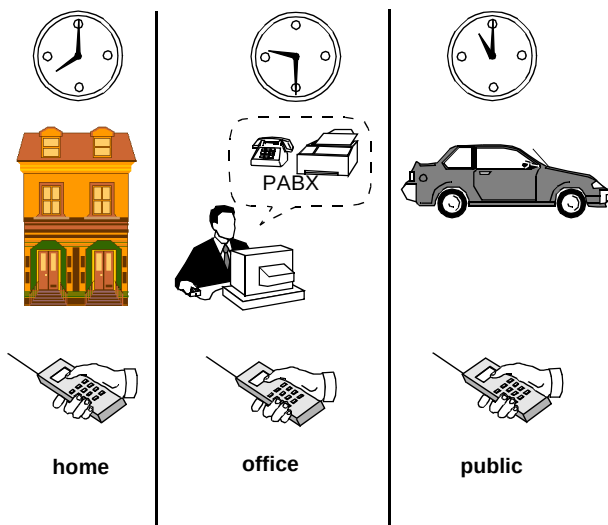


Fig 2 UMTS access domains.

For broadband to support UMTS it must be able to provide seamless service support across all access types. It therefore must be able to communicate across networks at a service level. This will require the visited (serving) network to use information such as location and subscriber profile downloaded from the terminal or home network to handle the service requests of its newly arrived subscriber. Broadband must also provide the capabilities to support other consequences of the 'inbound roamer' and mobile nature of the terminals such as handover, authentication and security.

As well as catering for real-time connection-oriented communications such as telephony and interactive video, UMTS will support connectionless/packet-type services to cater for less delay-critical data applications such as the Internet/intranets. Within the GSM community, support of connectionless communications is being developed via GPRS (Generalised Packet Radio Service) [3] which uses a modified Internet protocol (IP) and packet-transport mechanism to support direct connections to packet data networks. The handling of mobility management aspects has been achieved by interaction with traditional GSM network elements.

Although GSM supports the mobility requirements and the forerunner to VHE (via CAMEL), the system remains constrained by the (64 kbit/s) narrowband ISDN (N-ISDN) technology within the core network and the radio interface used. UMTS is expected to overcome these restrictions by enabling a flexible approach to data communications to enable connection and connectionless (packet) data capabilities including the support of asymmetrical data on the uplink and downlink, and variable data rates which allow applications to be added and dropped during communications [4]. For this reason broadband has been considered to support the variable asymmetric transport requirements. Broadband must therefore progress its developments from the current leased-line PVC-managed virtual circuits to include short duration SVCs. This will allow public access to the variable bit rates predicted for UMTS without paying the costs of 'business leased line'

rental. At the same time there must be inter-communication between the SVC's control and resource allocation functionality, service control, to cater for users' service requirements and preferences (and the subscription which determines if the user can request such capability), and mobility aspects such as handover and path re-routeing. If this cross-interrogation is not available, an illegal request for resources may be allocated, or one which the resources of the environment cannot support; a typical example would be a request for a high bit rate service in a low bit rate, high-mobility domain. With the developments of IP in the community, broadband cannot propose to be a UMTS system if it does not support common intelligence with IP. At present IP and broadband are seen as two separate entities; however, it is envisaged that as telecommunications becomes more multimedia based, there will be a greater need for a combined system providing both time-critical and delay-tolerant transmission capability. Only a convergence of technologies supporting both broadband and IP will enable UMTS to support the harmonised execution of mixed media communications envisaged.

2. The phased introduction of UMTS

Due to the widespread adoption of GSM within Europe and world-wide, ETSI has adopted a phased approach to UMTS to ensure a smooth transition for both operators and end users from second to third generation systems. Table 1 provides the typical thinking within the three phases adopted.

The initial usage of UMTS frequencies and the developing radio interface is made within phase 1. Phase 1 introduces the connection of the new UMTS terrestrial radio access (UTRA) to the current 64 kbit/s and GPRS infrastructure. This will provide the radio interface requirements to fully support UMTS, including asymmetric transmission, variable bit rate allocation and 'requestable' quality of service (QoS) communications, although in reality it will be limited due to the reliance on the legacy system to which it connects. Despite not actually being part of the UMTS development, the initial phase (termed phase 0) is considered as 'conditioning the market' towards UMTS via the application of contemporary DECT/GSM/DCS 1800 technology and 'peripheral' intelligent terminal based solutions as used in IP networks. Phase 2 is the true target scenario and will offer the full service capability. This will require significant changes within the network to fully exploit not only the UMTS radio access but also traditional 'fixed' access such as home and private LANs.

3. Radio interface requirements

The radio requirements for UMTS are critical in ensuring optimum use of the limited radio resource in providing mobile multimedia. The European UMTS forum has estimated that for the type and quantity of services envisaged for UMTS, a total 500 MHz of spectrum is required. Within Europe, UMTS currently only has 2×60 MHz of spectrum allocated for public terrestrial use. Hence there is a need to acquire more spectrum in which to operate, and/or find more efficient means of utilising the spectrum.

Table 1 The phased approach for UMTS.

UMTS phase	Service/customer capability	Network capability and radio spectrum utilised
Phase 0	GSM phase 1, 2 and some 2+ service set (voice and data services, connection and connectionless, IN service control, world-wide mobility via roaming, data rates up to 64 kbit/s). Seamless services (handover and location management). Simple terminal based (menu) control of supplementary services and features.	GSM radio access at 900, 1800 and 1900 MHz. N-ISDN based core network, plus an IP based network as overlay for connectionless data via GPRS. IN integration through CAMEL into connection mode network for both home and visited network based mobiles.
Phase 1	As above plus data rates up to 144 kbit/s via N $\times$ 64 kbit/s communications. High-quality speech (GSM EFR codec), service management via flexible user control interface. Roaming between UMTS, UTRA network and GSM.	Limited coverage from new UMTS radio interface system, interworking into existing IP and N-ISDN based core networks, mobility management as per GSM. Maintained use of GSM 900, 1800/1900 for mass coverage. Dual mode GSM/UMTS terminal.
Phase 2	Full UMTS requirements. As above plus simplified service control, virtual home environment (VHE) with automated roaming, home business and wide area usage. Support for full variable bit rate multimedia communications, alternate addressing mechanisms, and service adaptation.	First integrated fixed/mobile network. Access network is UMTS radio access system. Core network is connectionless based transmission supporting connection (real time) and connectionless traffic. Intelligence linkage to manage customer service profile, preferences and limitations, etc, satellite and terrestrial access to network.

The UTRA solution strives to satisfy the following key objectives :

- to provide a common wireless interface for roaming across all terrestrial environments,
- to provide a flexibly structured radio interface that can adapt to cater for the range of existing and future (as yet unknown) services — this includes a capability to support high bit rates,
- to make efficient use of available radio spectrum.

A range of radio environments have been identified in which the UTRA can operate, including indoor (residential and business) and outdoor (urban and rural). Each of these environments has differing radio propagation characteristics that influence parameters of the UTRA. The constraints that these impose are summarised in Table 2.

Table 2 The phased approach for UMTS.

Environment	Maximum user bit rate	Maximum user speed
Indoor (low range)	2 Mbit/s	10 km/h (6 miles/h)
Outdoor (urban)	384 kbit/s	120 km/h (75 miles/h)
Outdoor (rural)	144 kbit/s	500 km/h (311 miles/h)

The radio interface will be able to achieve full dynamic variable bit rate throughout the call (up to the maximum rates specified for the different domains in Table 2). To achieve this the channel QoS attributes can be negotiated at call set-up and adjusted throughout the call to suit the specific service required. The attributes that can be varied are:

- variable bit rate,
- variable delivery delay,
- variable duplex symmetry — it is probable that future typical applications will transmit larger volumes of data in one direction than the other (e.g. Internet browsing — high bit rate downlink but minimal rate uplink), and therefore to maximise efficiency in handling such traffic the radio interface should allow asymmetrical connections to be set up,
- variable protection levels to accommodate a range of acceptable bit error rate.

The requests to vary these attributes will be balanced by modification of the link according to quality, network load, and radio conditions, in an attempt to optimise the link in the different environments. This highlights the need for network and access communication and QoS negotiation for all resource requests.

4. Currently proposed radio technologies

The standardisation of UTRA is currently being undertaken within ETSI's SMG2 group and is limited to the terrestrial (i.e. excluding satellite) components of the radio interface. This is ongoing and a time schedule exists to choose the final UTRA technology, which will stem from a combination of the following proposals. The milestones in the decision process are:

- June 1997 — agreed UTRA requirements and evaluation procedure,
- December 1997 — selection of one UTRA concept,
- June 1998 — definition of the key technical aspects of the selected UTRA.

Several proposals have been presented to SMG2 [5], originating from a range of manufacturers and operators. The key difference between proposals is the multiple access technique chosen, this being the factor that determines most of the other system characteristics.

The three fundamental access techniques are FDMA (frequency division multiple access), TDMA (time division multiple access) and CDMA (code division multiple access). All proposals are based on hybrids of these technologies, with the proposed radio access techniques that show similarities being collected into five concept groups to ensure co-operative development.

These concept groups are currently evaluating the technologies against the set of requirements. The final decision as to which UTRA concept is finally chosen, will be based upon their output.

UTRA highlights the European position for a third generation mobile system. Within other world regions different standards are being developed. It looks increasingly likely that the ITU will not necessarily standardise a single solution, but implement a framework to allow roaming between the different regional solutions (known as the 'family concept'). One suggestion is to standardise a common global broadcast signalling channel which would indicate to a mobile station which air interface technology to operate in a particular environment.

This common signalling channel could also be useful in each region/environment to indicate which portions of the spectrum should be used and which should be avoided. In order to accommodate regional variations in available spectrum, the radio interface should be capable of operating in any feasible frequency band that becomes available.

5. Network requirements

Due to the multiservice and capability nature of UMTS it is essential to select the correct technologies within the network to provide the required functionality. High-level analysis of network intelligence requirements to support mobility and service control has been presented previously [6]. The underlying issue of the transport technology beneath the network still remains.

When considering the information to be transferred by UMTS, a rough separation can be made in the data types' characteristics, which are summarised below.

- Message

Connectionless-'data', time-delay tolerant, but requiring low BER — this is generally a fixed-length single packet, for handshake type operations, as seen for signalling packets.

- File

Connectionless-'data', time-delay tolerant, but requiring low BER — generally this is variable in length, e.g. WWW Internet page download, file transfer.

- Stream

Connection-oriented — this can be further separated into two sub-groups:

- delay intolerant, such as 'speech' and 'videophone' like applications, normally characterised by two-way (duplex) interactive, which is time and delay intolerant, but is more flexible in its BER demand,

- delay 'some' tolerance, such as in video broadcast where the terminal may be storing the data, but there is no catastrophe if small time delays occur.

To support the carriage of the different user information (while still maintaining communications in the hostile mobile environment), a combination of bit transport, call and connection control, charging, service and mobility control is required. The application of a number of contemporary technologies has been considered to satisfy the UMTS requirements, divided into circuit switched, and packet and variable rate switched.

- Circuit switched (64 kbit/s):

- PSTN/ISDN/GSM,

- 64-kbit/s technology,

- fixed bit rates ($N \times 64$ kbit/s),

- linkage to 'IN' and 'mobility',

- potential B-ISDN evolution.

- Packet and variable rate switched:

- continuous and discontinuous,

- ATM, GPRS, IP and mobile IP,

- mainly bit transport,

- flexible, variable rate and QoS control possible,

- no external 'control',

- potential mobility via mobile IP.

Of the technologies considered the former 64-kbit/s path is not a realistic solution due to its rigid bit rate and transport capability; however, it does support interaction with external intelligence. For the latter technologies the packet and variable bit rate mechanisms provide good transport capability, but are currently inflexible with limited interactions to external intelligence.

Current thinking to satisfy the UMTS network transport and control requirements involves both adopting ATM/IP transport and including links into intelligence for service control and mobility management. ATM has been selected due to its switching speed and potential QoS parameters and IP for its current widespread use and simple interworking with legacy systems. This will satisfy the flexible user bandwidth (high and low bit rates), asymmetric data services, multimedia, variable QoS and connectionless/connection-oriented requirements, while also enabling the adoption of B-ISDN (Q.298X) signalling technology and the developing DAVIC concepts [7] for call/media control.

6. B-ISDN and IP developments

In the development of broadband, the following stages are seen as major steps to fully support the UMTS requirements. Many of the following proposals are already under consideration or development for other purposes; however, in order to support UMTS it is essential that the mobile-specific deltas are included.

The introduction of ATM into the network is via leased line permanent virtual circuits (PVCs). These circuits are permanently nailed up, set up by external management procedures and on a leased line basis. This is a permanent connection providing high-bandwidth capabilities with no possibility for service, call control or service interaction. The next step is the introduction of soft PVCs, which support signalling to dynamically re-route PVCs should failure to links or switches occur. Not only does this make the network more resilient, it also introduces a signalling mechanism leading to the next obvious technology step of switched virtual circuits (SVCs) providing dynamic

allocation of virtual circuits on demand. SVCs support call control capabilities (in a similar way to N-ISDN's 64-kbit/s bearers) by allowing variable bit rate bearer allocation. The separation of bearer and call control allowing several bearers to be allocated to a single session is the next fundamental step. This not only supports dynamic variable bit rate allocation throughout the call but also provides a potential solution for mobile handover by allowing legs to be added and dropped within a 'communications session' when required.

With developments in service control to provide mobility procedures, the ability for the intelligence to interact with the ATM transport to add and drop bearers will be essential to physically support handover and re-routing procedures. Handover and variable bit rate allocation will only be efficiently supported in ATM if there is a separation from bearer control and call control. If there is no separation between bearer and call control, complete end-to-end 'calls' will have to be built up to add or drop calls, which includes the handing over of the terminal to another access point. This obviously has serious impact on the resources of the system.

IP's development for UMTS builds upon the introduction of some intelligence within the network to support the mobility requirements of mobile IP. To be fully compliant with the UMTS goal, this mobile information needs to be accessible from other networks and to be modifiable to update the information present in its database. Thus any other network can have access to the location information of a terminal without needing to send the packet to the terminal's network only to be forwarded out again.

Another technological hurdle at present is the convergence of ATM and IP. Considering that IP is a connectionless mechanism and ATM a connection-oriented one, there are immediate problems seen in the transport of information and how these two methods can be converged. What UMTS implies is that a common method of 'control' or linkage of information will be available to both mechanisms. This means there will be control information, such as mobile information, subscriber preferences and limitations, and security information, which relate to both transport choices for UMTS. It also implies that a single common system will control mobile procedures, call control and resource allocation and modify parameters throughout the session.

For ATM to support a connectionless non-resource-allocating mechanism, such as IP, a route relationship needs to be established (without the allocation of any set routes or resources until the packet is being transmitted) and then closed down once transmission of the packet is completed. On the other hand, if IP wished to provide a circuit-switched

type service, it would have to introduce call control and resource allocation procedures throughout, clearly going against IP original thinking. It therefore seems unlikely that solely one or the other technology will exist in isolation; however, there are more optimistic views on common databases and intelligent servers influencing/controlling the lower transport mechanisms.

What has been presented previously are the initial high-level thoughts on the ability of broadband and IP (with developments) to support UMTS. The adoption both of this evolved ATM and of IP with links to intelligence still requires development and more detailed study to consider the interworking functionality required to contemporary systems, as do the more detailed mobility, security and radio-related issues.

7. The network architecture for UMTS

The vision of UMTS is of convergence between fixed and mobile networks. The core network will support both wired and wireless access. Figure 3 illustrates the conceptual architecture required for UMTS.

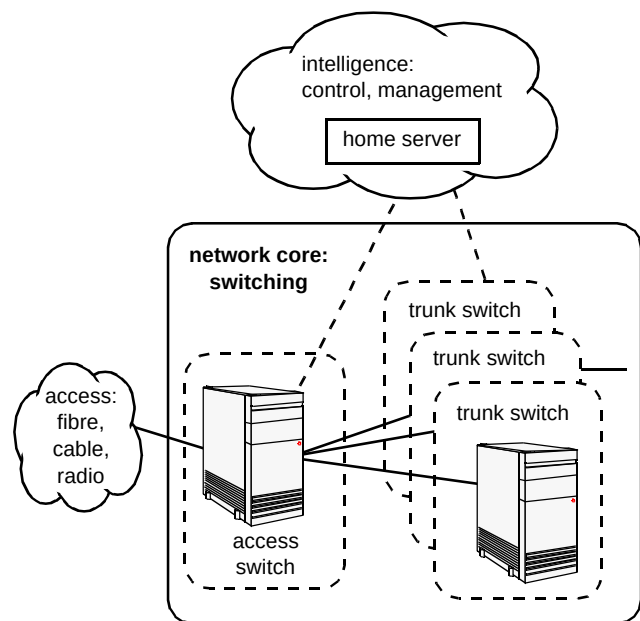


Fig 3 Conceptual architecture for UMTS.

The three main components for UMTS (or any telecommunications system) are access (radio or wired), network transport/switching and overlying service and mobility control.

Based upon the overall architecture requirements and the analysis of transport and intelligence options, a

conceptual functional network architecture has been developed as shown in Fig 4.

The main areas depicted in Fig 4 can be separated out into the home server, the UMTS server, transport, access and the terminal. These are detailed below.

- Home server

This is likely to be a single entity containing the subscriber's profile, which consists of the complete list of services subscribed to, and their constraints, as well as the particular preferences of the user. It also contains the most recent location information and authentication/security information and hence is interrogated for terminal terminating calls and initial registration when the terminal is requesting attachment to the network.

- UMTS server

This is seen as a distributed intelligent layer, although depicted for ease as a single entity. The UMTS server controls the resource allocation, call control and bearer allocation. For mobile terminals the UMTS server will control the mobility procedures and associated resource allocation. As a constant interaction with the service profile is required to enable the VHE for UMTS, a copy of the relevant parts of the subscriber's

profile is present, downloaded from the home server. Information from the access environment is also essential to identify which resources are available and under what restrictions.

- Transport

The transport system encompasses two mechanisms, packet switched and cell switched (connection oriented). It is assumed all mobility, call and service control intelligence will be taken out of the switch/router and decisions on routing and resource allocation are to be carried out in the distributed UMTS server.

- Access

Access connects the network to the user. The transfer of information from the access type to the UMTS server is essential to inform it of the current resource status of the access and its capabilities/limitations.

- Terminal

The terminal will have much greater contact with both the network and access systems; communication is likely to be via an Internet front end.

Developments for this architecture include the adoption of the IP switching/router and ATM-based transport in parallel with links to the distributed intelligence processing (home/serving division).

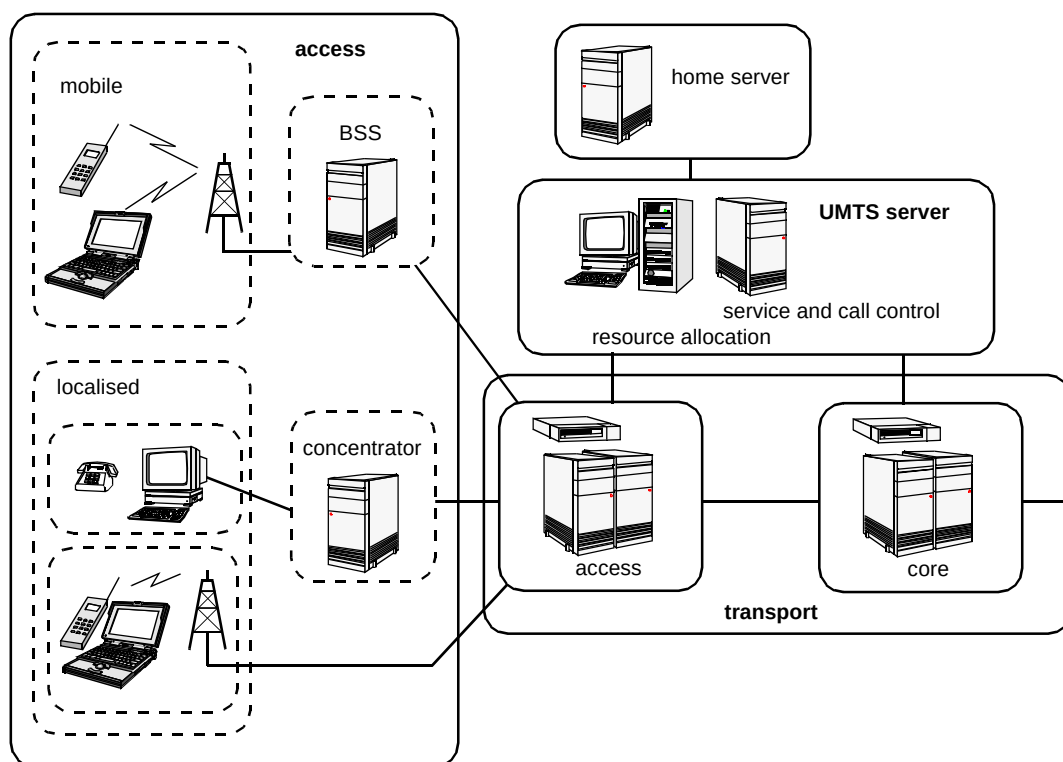


Fig 4 Network architecture for UMTS.

8. Key network operations for UMTS

To illustrate the operation of the network this section introduces a number of typical network operations. Figure 5 illustrates initial registration with UMTS and involves interaction between the terminal and the network intelligence (both home and serving) to check and transfer the user's relevant service information (including information such as VHE capability) into the serving network's intelligence. As shown, one potential technology which can be applied to such activity is the developing mobile IP; this would remove the need for a dedicated channel on registration and so increase efficiency for signalling over the air interface.

Figure 6 illustrates the procedures to set up mobile-originated connectionless (packet) communications.

The operation involves the terminal interacting with the network intelligence to set up a 'session'. The session defines the requirements for the communications. The intelligence verifies whether the request is legal in relation to the subscriber's profile and communicates with the packet mechanism of the network and access nodes. The network intelligence will configure the packet network's routing tables to enable the correct packet route. The 'session' will be maintained as the mobile moves, with the network path being re-configured accordingly, with data 'cached' and re-routed as required for 'handover'.

A similar operation is performed for mobile-terminated connectionless (packet) set-up as shown in Fig 7.

It should be noted that for the mobile-terminated connectionless (packet) set-up the home server is involved to configure the packet routing to the serving network (to invoke a terminal-initiated 'session set-up'); this also involves a check on the called mobile's subscribed services and capability. Once the (mobile-originated or terminated) path has been set up (via the session being in place), the actual transmission path is open to convey the packets of data. The path adopted is illustrated via the dashed lines in Fig 8.

If the path needs modification the mobile/server will interrupt communications and modify the routing configuration as illustrated in Fig 6.

A very similar operation is applied to connection-oriented communications, the only difference being the 'session' is on a 'per-call' basis. The server will interact with the ATM switching to operate services and to support path modification for handover and re-routing, etc, by dropping or adding bearers accordingly.

To enable UMTS to cater for variable bit rate multimedia communications, it is essential that the network can modify, add and drop communications components 'in call'. This is achieved by modification of the parameters discussed within the 'session' as shown in Fig 9.

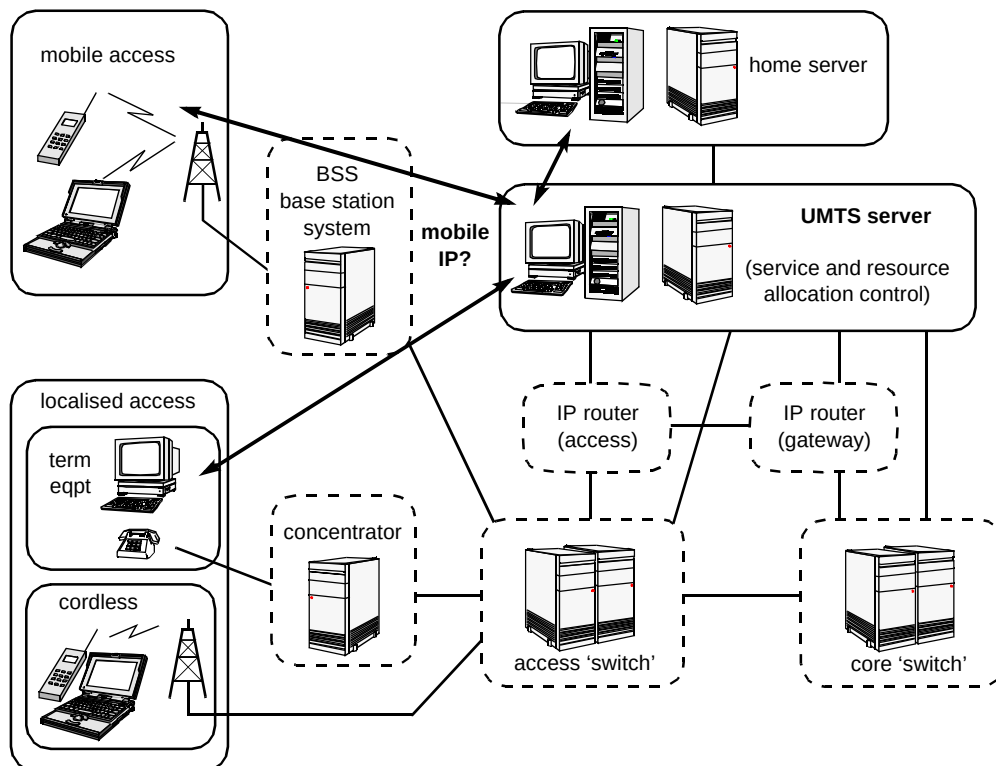


Fig 5 UMTS registration.

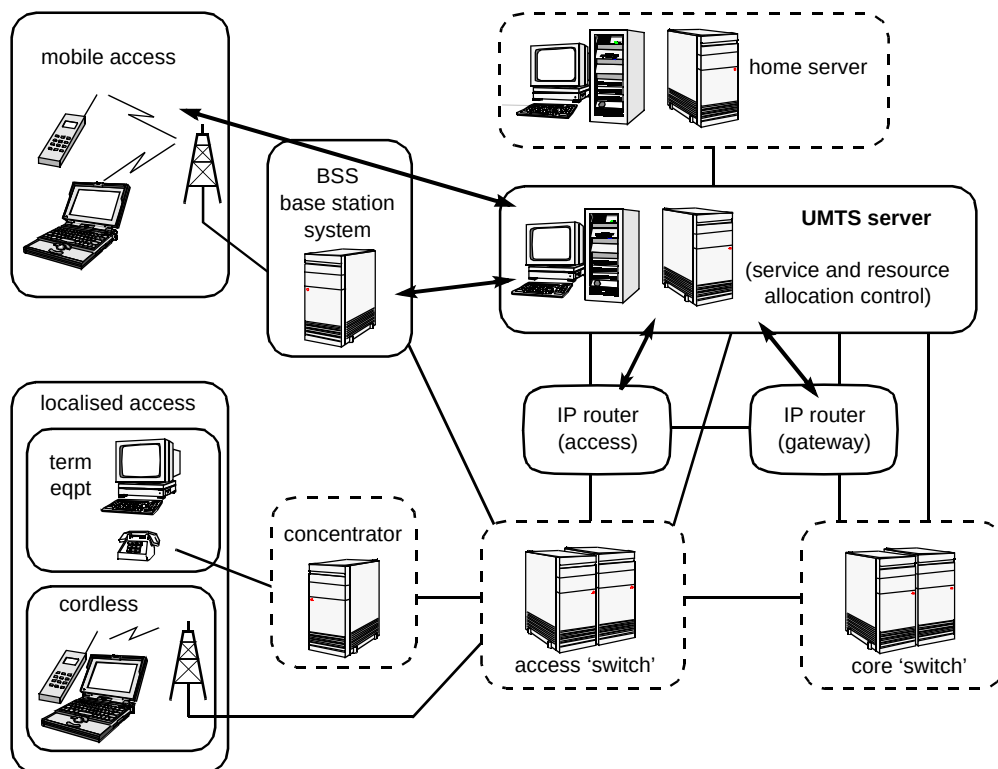


Fig 6 Mobile-originated connectionless (packet) set-up.

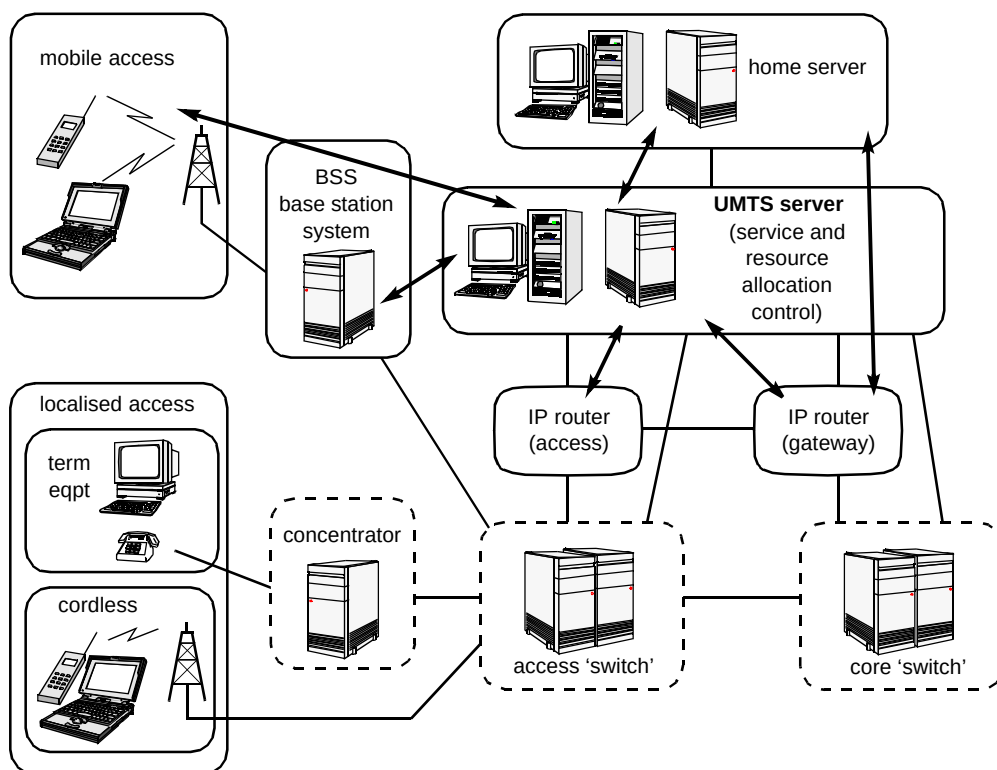


Fig 7 Mobile-terminated connectionless (packet) set-up.

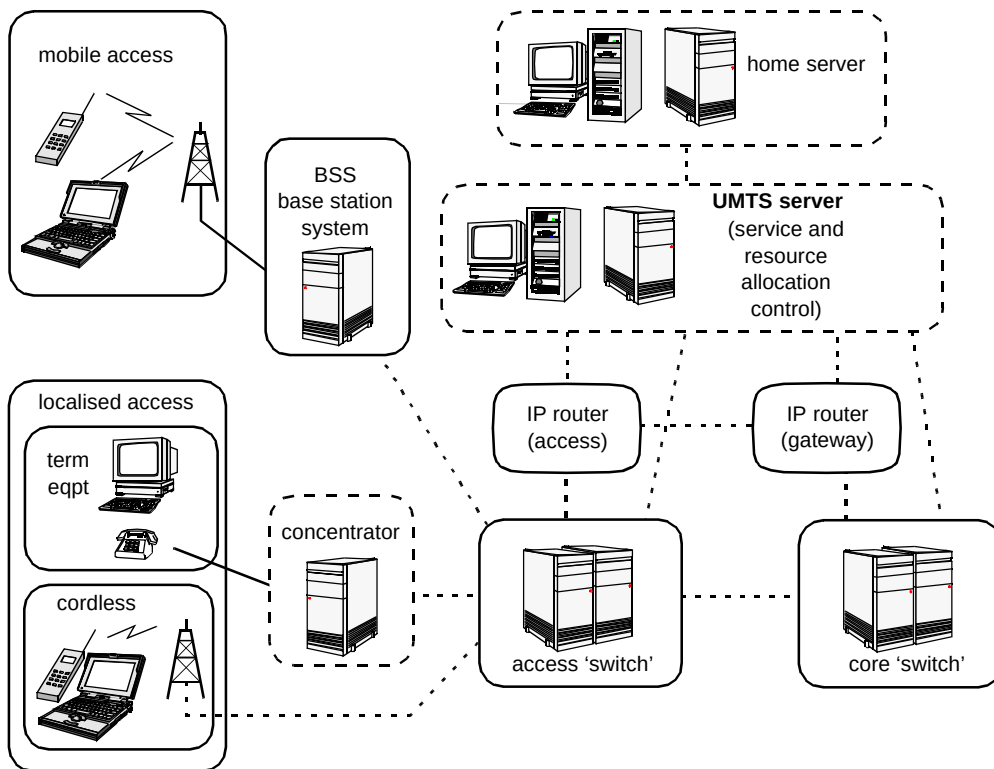


Fig 8 Connectionless (packet) transmission.

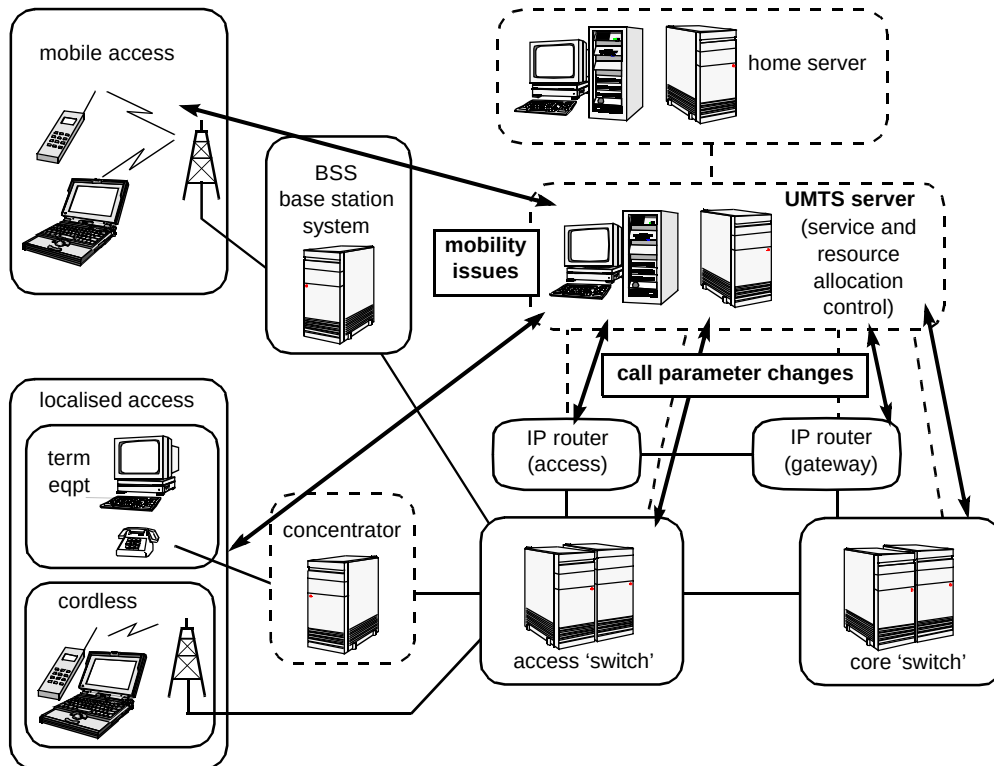


Fig 9 Session modification.

Changes in the session can either occur due to changes in the user/application requirements and are termed 'application-initiated', or be caused by the network and are termed 'network-causal'. These are typically due to the inability of the network to continue to support the communications in their present form — hence a re-negotiation of the session parameters is required. Network-causal changes can be further separated into mobility aspects due to the mobile environments (e.g. handover, location or user service constraints), and network resource aspects which allow more users to have access to the (limited) network radio resource via the lowering of priority settings for transmission.

9. Conclusions

From a high-level perspective, the major development UMTS offers is the provision of multi-environment, multimedia mobile communications. The network technology to fulfil UMTS requirements still needs evolution and enhancement, the major development being the linkage of connectionless (packet) and connection-oriented (real-time) capability and then enabling this technology to interact with network intelligence.

For broadband and IP to fully support UMTS, developments must encompass the following points:

- develop broadband interaction with IN,
- develop mobility and service control within intelligence on the transport mechanism,
- standardise completely the separation of call control and bearer control to allow for the dynamic variable bit rate and handover capabilities required,
- develop IP interactions with IN,
- develop methods of linking intelligence across IP and broadband systems.
- develop how both IP and ATM transmission technologies will support common mobility control via the intelligence,
- standardise capability to allow for inter-vendor and inter-network operability on a capability level to support the UMTS requirements such as VHE.

10. Time lines

The following time lines are separated into the radio requirements that have been identified according to the SMG workplan, and the network requirements that would be necessary to support this added functionality.

10.1 Radio requirements

The following is a summary of the predicted time line according to the SMG workplan:

- 1997 — radio access technology chosen for the 2-GHz air interface with up to 2-Mbit/s support,
- 1998 — basic standardisation on chosen radio access technology completed,
- 1999 — testbed trials,
- 2001 — availability of equipment from vendors,
- 2002 — the UMTS air interface is introduced with interworking between UMTS and the legacy systems of that time (GSM 900/1800); this would allow up to 2 Mbit/s enabling communications via blocks of $N \times 64$ kbit/s.

10.2 Network requirements

To meet the above requirements, the following would need to be achieved within the times specified.

- 2000 — standardisation complete covering the following areas:
 - switched virtual circuit definition,
 - subscriber profile storage,
 - inter-service communication to provide personal mobility and limited VHE,
 - ATM separation of call and bearer control providing dynamic variable bit rate allocation,
 - broadband support of mobility procedures for terminal mobility,
 - interaction with IP 'intelligence' to provide common control of mobile and service interaction in the network,
- 2002 — test bed trials,
- 2003 — availability of equipment from vendors,
- 2005 — implementation of UMTS within the network:
 - evolution of a ubiquitous transport mechanism, to provide efficient carriage of both connectionless and connection-oriented call types within the network,

— development of full intelligence interaction to provide service and mobility management control to allow handover and operation in all access domains including the support of service adaptation and full dynamic broadband rate allocation.

With these capabilities UMTS will be the common network platform to support fixed and mobile telecommunications into the next millennium.

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He is currently providing the technical lead for BT's network architecture project on UMTS, has participated in EC projects on mobile and IN networks, and has developed IN and GSM network aspects for Cellnet's live GSM network in the UK.



Steve Dutnall joined BT Laboratories in October 1996, after completing a 6 month project placement at BT as part of his MSc. He has worked on the impact of IN on GSM (CAMEL) and UMTS.

He is currently contributing to BT's projects on UMTS, focusing on the transport mechanisms and network evolution for this third generation system.



Mark Dando has been working at BT Laboratories since 1995. He currently co-ordinates the radio aspects and advises on the industry structure of UMTS.

His background also involves investigation of different radio planning strategies for Cellnet's live GSM network.



Alan Clapton joined BT in 1967 as an apprentice at the high frequency radio station at Leafield near Oxford, and subsequently graduated with a BSc honours degree in Electrical Engineering. His career has included aeronautical and maritime radio developments as well as GSM service and network developments. Recent work has included supporting Cellnet on GSM data and value added services as well as intelligent networks.

In parallel he has led the BT Laboratories team developing UMTS technical aspects.



Pedro Serna graduated from Cantabria University, Spain, in 1994 and gained a diploma thesis from Ulm University Germany on the investigation of CDMA systems in 1995.

He joined BT Laboratories in November 1996 and has been assessing and analysing radio technologies for UMTS.